- **Accessibility**: The ability to easily access, read, write to, and use the application in real time
- **Governance**: The ability to govern vector storage with the same access controls as the original sources used to create the vector embeddings and models
- **Integration**: The ability to easily integrate with current market technologies and eliminate time spent stitching technologies and solutions together

Using embeddings and vector search is a powerful way to improve a variety of ML projects. There are many uses for vector databases, the most common of which are as follows:

- **RAG systems**: Vector search facilitates efficient data retrieval, which is then used to augment a **Large Language Model** (**LLM**)'s response. Augmenting an LLM with results from vector search leads to more accurate chatbot responses and minimizes errors such as hallucinations in LLM outputs.
- **Recommendation systems**: E-commerce and streaming platforms use vector search for efficient nearest-neighbor searches, matching user behavior with relevant suggestions.
- **Image and video recognition**: Vector search facilitates quick searches for similar features in images and videos.
- **Bioinformatics**: Vector search can be applied to tasks such as DNA sequence alignment and protein structure similarity search to improve clinical research.

Enhancing data retrieval with Databricks Vector Search

Databricks VS is transforming how we refine and retrieve data for LLMs. Functioning as a serverless similarity search engine, VS enables the storage of vector embeddings and metadata in a dedicated vector database. Through VS, you can generate dynamic vector search indices from **Delta** tables overseen by Unity Catalog. Using a straightforward API, you can retrieve the most similar vectors through queries.

Here are some of Databricks VS's key benefits:

- **Seamless integration**: VS works harmoniously within Databricks' ecosystem, particularly Delta tables. This integration ensures that your data is always up to date, making it model-ready for ML applications. With VS, you can create a vector search index from a source Delta table and set the index to sync when the source table is updated.
- **Streamlined operations**: VS significantly simplifies operational complexity by eliminating the need to manage third-party vector databases. VS runs on serverless compute, meaning Databricks handles the infrastructure management for you.
- **Enhanced scalability**: Unlike standalone vector libraries, VS offers unparalleled scalability. VS handles large-scale data effortlessly, scaling automatically to meet the demands of your data and query load. This scalability is crucial for organizations with vast amounts of data and complex search requirements.